Bheri Sai Likith Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.79
Sri Chaitanya	Class XII (Board of Intermediate Education Andhra Pradesh)	2024	96.60%
Sri Chaitanya	Class X (Board of Secondary Education Andhra Pradesh)	2022	97.50%

SKILLS

Skills : Data Structures & Algorithms, Machine Learning, Deep Learning.

Programming Languages : Python, C++, C#

Libraries, Frameworks &Tools : NumPy, Pandas, Matplotlib, Scikit-Learn, PyTorch, FastAPI, ChromaDB, Git, GitHub

PROJECTS

Codebase Intelligence | *Self Project* | FastAPI, ChromaDB, SentenceTransformers, Gemini API

- Built a RAG-powered **API** that answers natural language questions about any GitHub repository by cloning, chunking, and semantically indexing Python and Jupyter Notebook files.
- Implemented **AST-based chunking** to split code at function and class boundaries, improving retrieval precision over naive line/character splitting.
- Designed a batched **AI summarization** pipeline using AI API to generate natural language descriptions per code chunk, bridging the semantic gap between code and natural language queries.
- Generated structured **repository overviews** including file tree, key module descriptions, tech stack, and entry points using a single context-aware AI prompt.

Machine Learning Algorithms & Mini ML Library | *Python*

- Developed a modular, **scikit-learn-inspired machine learning library** in Python, implementing reusable preprocessing, model selection utilities, and multiple ML algorithms with a unified fit/predict.
- Implemented **classical machine learning algorithms from scratch**, including K-Means Clustering, Gaussian Mixture Models (EM), Logistic Regression.
- Implemented deep learning models from scratch using NumPy and PyTorch, including MLPs, CNNs, RNNs, LSTMs, and Transformer Encoders.

Rowmance | *Glitch, the Gaming Club of IIT Hyderabad* | Unity Engine, C#

- A Unity-based co-op game requiring two players to coordinate boat navigation avoiding obstacles and completing the map.
- Implemented gameplay systems including whirlpools, jump pads, **power-ups.**
- Designed and optimized modular gameplay scripts, improving code maintainability.

MILAN 2025 Minigame Collection | *Glitch, the Gaming Club of IIT Hyderabad* | Unity Engine, C#

- Developed one of the five **2D Unity** minigames for **MILAN 2025,** IIT Hyderabad's annual inter-hostel competition.
- Developed a 2D endless mountain-climbing platformer in **Unity (C#)** with procedural level generation.
- Implemented player mechanics including **wall jump, double jump, coyote time** for responsive gameplay.
- Designed an optimized spawning system using **object pooling** for platforms, obstacles, and backgrounds to support smooth infinite gameplay.

RELEVANT COURSES

Artificial Intelligence and Machine Learning : Foundations of Machine Learning, Deep Learning, AI for Humanity, Programming for AI, Introduction to Modern AI, Intro to Classical AI, Probability and Random Variables, Matrix Theory

Computer Science : Data structures and Applications, **Algorithms**, Data Base Management System - 1, Introduction to Programming.

Mathematics : Optimization, Numerical Analysis, Differential Equations, Vector Calculus, Calculus, Discrete Math for Computer Science.

POSITIONS OF RESPONSIBILITY

Core Member, Game Developer - Glitch, The Game Development Club, IIT Hyderabad

- Developed an endless platformer game independently, handling core mechanics and gameplay systems for MILAN
- Collaborated on a two-player cooperative game where each player controlled one paddle of a shared boat, requiring synchronized input to navigate
- Supported game development workshops by answering participant questions and clarifying programming and Unity concepts.

SCHOLASTIC ACHIEVEMENTS

- Solved **150+ Leetcode** questions, developing understanding on Data Structures and Algorithms.

EXTRACURRICULAR

As part of **National Service Scheme(NSS),** participated in community awareness campaigns and walkathons promoting health, social responsibility, and public welfare.